Appendix II
Mathematical Model and Training Protocol

Supplementary Material for the IEEE Internet of Things Journal Submission

1 Purpose of This Appendix

This appendix expands the mathematical definitions that are compressed in the eight-page manuscript. It clarifies the multilayer graph representation, the relation-specific adjacency matrices, the GCN update, the attention fusion mechanism, the optimization objective, and the computational interpretation.

2 Multilayer RPL Graph

At each observation window, the parsed Cooja trace is converted into

$$\mathcal{G}_s = (V, \{A_s^{(\ell)}\}_{\ell \in \mathcal{L}}, X_s, Y_s), \qquad \mathcal{L} = \{r, q, t, a\}. \tag{1}$$

The set V contains the same physical motes in every relation layer. The four layers encode different evidence channels: routing (r), link-quality (q), temporal behavior (t), and trust/anomaly similarity (a). The feature matrix is $X_s \in \mathbb{R}^{N \times F}$ and the label vector is $Y_s \in \{0,1\}^N$.

The same physical node can be viewed as a set of layer replicas (v_i, ℓ). The conceptual supra-adjacency is

$$\mathcal{A}_{sup} = \begin{bmatrix} A^{(r)} & \lambda I & \lambda I & \lambda I \\ \lambda I & A^{(q)} & \lambda I & \lambda I \\ \lambda I & \lambda I & A^{(t)} & \lambda I \\ \lambda I & \lambda I & \lambda I & A^{(a)} \end{bmatrix}. \tag{2}$$

The implementation does not need to materialize this large block matrix. Instead, it computes relation-specific convolutions on the diagonal blocks and performs cross-layer coupling through fusion over the same node replicas. This makes the method easier to ablate because each RPL relation remains identifiable.

3 Node Features

For node v_i, the observable feature vector is

$$x_i = [\tilde{i}, \tilde{R}_i, \tilde{G}_i, \tilde{S}_i, \tilde{M}_i, \tilde{N}_i, \tilde{D}_i, \tilde{T}_i, \tilde{P}_i], \tag{3}$$

where each component is standardized within the snapshot. Standardization avoids letting high-magnitude routing counters dominate lower-magnitude link and delay fields. No advertised-rank decrement or DRA audit hook is included in x_i.

4 Routing Layer

Let $p(i)$ be the preferred parent of node v_i. The routing layer is

$$A_{ij}^{(r)} = \begin{cases} 1, & p(i) = j \text{ or } p(j) = i, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

This layer is important because DRA directly targets parent selection by making the attacker appear closer to the root.

5 Link-Quality Layer

The link-quality layer uses the same observed parent-child support but weights edges according to the parent metric:

$$A_{ij}^{(q)} = \begin{cases} \left(\max\{M_i/M_0, 1\}\right)^{-1}, & p(i) = j \text{ or } p(j) = i, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Lower parent metric values imply stronger weights. This layer lets the model distinguish a structurally attractive parent from a reliable parent.

6 Temporal Layer

Let

$$z_i^{(t)} = [D_i, \Delta p_i, \Delta R_i], \tag{6}$$

where D_i is delay, Δp_i is parent-switching behavior, and ΔR_i is rank-change behavior in the observation window. Temporal similarity is

$$S_{ij}^{(t)} = \exp\left(-\frac{\|z_i^{(t)} - z_j^{(t)}\|_2^2}{\sigma_t^2}\right). \tag{7}$$

Edges are retained when this similarity is among the strongest temporal relations. This layer is not an attack label; it is a behavioral neighborhood showing which nodes exhibit similar dynamics.

7 Trust/Anomaly Layer

The trust/anomaly descriptor is

$$b_i = [a_i^{(R)}, a_i^{(M)}, 1 - P_i], \tag{8}$$

where $a_i^{(R)}$ is a low-rank anomaly score, $a_i^{(M)}$ is parent-metric deviation, and $1 - P_i$ is packet-loss behavior. The corresponding similarity is

$$A_{ij}^{(a)} = \exp\left(-\frac{\|b_i - b_j\|_2^2}{\sigma_a^2}\right). \tag{9}$$

This layer should be interpreted as suspicious-behavior similarity, not a direct maliciousness label. It gives the GCN a structured channel to compare nodes with similar abnormal patterns.

8 Layer-Specific GCN

For every layer ℓ, self-loops are added:

$$\tilde{A}^{(\ell)} = A^{(\ell)} + I. \tag{10}$$

The normalized adjacency is

$$\hat{A}^{(\ell)} = (\tilde{D}^{(\ell)})^{-1/2}\tilde{A}^{(\ell)}(\tilde{D}^{(\ell)})^{-1/2}, \qquad \tilde{D}_{ii}^{(\ell)} = \sum_j \tilde{A}_{ij}^{(\ell)}. \tag{11}$$

A one-hop relation-specific convolution is

$$H^{(\ell)} = \phi(\hat{A}^{(\ell)} X W^{(\ell)} + b^{(\ell)}). \tag{12}$$

The operation means that node v_i updates its representation by combining its own features with the features of neighbors connected under relation ℓ.

9 Attention Fusion

The non-attentive ML-GCN concatenates relation embeddings with fixed layer treatment. Attn-ML-GCN instead learns a scalar score θ_ℓ for each relation:

$$\alpha_\ell = \frac{\exp(\theta_\ell)}{\sum_{m\in\mathcal{L}} \exp(\theta_m)}. \tag{13}$$

The fused representation is

$$H = [\alpha_r H^{(r)} \| \alpha_q H^{(q)} \| \alpha_t H^{(t)} \| \alpha_a H^{(a)}]. \tag{14}$$

This formulation keeps the relation channels separated while still allowing the model to emphasize stronger evidence channels during training.

10 Node Classification and Loss

The classifier maps H_i to a two-class probability vector:

$$Z_i = \mathrm{softmax}(H_i W_o + b_o). \tag{15}$$

Weighted cross-entropy is used:

$$\mathcal{J}(\Theta) = -\sum_{i\in\mathcal{V}_{tr}} \sum_{c=0}^{1} \omega_c \mathbb{I}(y_i = c) \log Z_{i,c}. \tag{16}$$

The positive-class weight ω_1 compensates for the fact that malicious motes are the minority class at 10% and 20% attack ratios.

11 Threshold Selection

The model outputs probabilities, but the final detector needs a binary decision. The threshold τ is selected on validation seed 4:

$$\tau^\star = \arg\max_{\tau\in[0,1]} \mathrm{F1}_{val}(\tau). \tag{17}$$

The selected threshold is fixed before evaluating test seed 5. This avoids tuning on the test set.

12 Computational Cost

For sparse matrices, one forward pass has dominant cost

$$\mathcal{O}\left(\sum_{\ell\in\mathcal{L}} |E^{(\ell)}|F + |\mathcal{L}|NFd + N|\mathcal{L}|d\right). \tag{18}$$

The first term is sparse message passing, the second is feature projection, and the third is fusion. Attention adds only a small cost:

$$\mathcal{O}(|\mathcal{L}|Nd + |\mathcal{L}|). \tag{19}$$

13 Permutation Equivariance

For any node permutation matrix P, the model satisfies

$$f_\Theta(PX, \{P\hat{A}^{(\ell)}P^T\}_{\ell\in\mathcal{L}}) = Pf_\Theta(X, \{\hat{A}^{(\ell)}\}_{\ell\in\mathcal{L}}). \tag{20}$$

Thus, predictions are tied to graph structure and features rather than arbitrary node numbering.